Assignment 2

Computational Plasticity (SoSe25)

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Given Data

In this section, the parameters value for the material properties, that required by the author to calculate it based on the author's Immatrikulation Nummer, will be defined accordingly. To make the calculation detailed and transparent, each parameter will be derived step by step.

$$a = 4, \quad b = 6, \quad c = 8 \tag{1}$$

Parameter for 1st question:

- $E = 200 + (10 \times a) = 200 + (10 \times 4) = 240$ GPa

Parameter for 2nd question:

- $\sigma_y = 600 + (a \times 33) = 600 + (4 \times 33) = 600 + 132 = 732$ MPa
- $C_1 = 10000 + b \times 1666 = 10000 + 6 \times 1666 = 10000 + 9996 = 19996$ MPa
- $\gamma_1 = 20 + (3 \times c) = 20 + (3 \times 8) = 44$ MPa
- $C_2 = 3000 + (a \times 166) = 3000 + (4 \times 166) = 3000 + 664 = 3664$ MPa

Parameter for 3rd question:

- $\sigma_{ya} = 300 + (10 \times b) = 300 + (10 \times 6) = 300 + 60 = 360$ MPa

1 Introduction

For this second assignment, mechanical response analysis will be conducted using a simple squared geometry model with a circular inclusion in the center. Hence, the analysis will be divided into three tasks. The first task is mesh sensitivity analysis, where the main goal of this first task is to see the influence of variety of mesh sizes to the captured stress concentration generation along the defined line from the edge of the circular hole until the opposite edge of the plate. Then there's come the second task, where the focus will be to investigate the isotropic and kinematic hardening effect to its mechanical response. The last task is a subroutine implementation, where the objective is to point out the difference between the standard and UMAT results. Extracting some other information will be done as well in the third task.

In this following section, each question will systematically will be addressed in the assignment, providing detailed explanations, derivations, and relevant figures or tables to support the analysis. All calculations and results are based on the provided data and the author's unique identification number. Numerical analysis will be performed using Abaqus CAE with an appropriate boundary condition and settings.

2 Set up of the Model

For the model setup, 2D planar-deformable shell-model will be used for the plate with the specified geometry, boundary condition, material properties, and mesh.

2.1 Geometry and Boundary Conditions

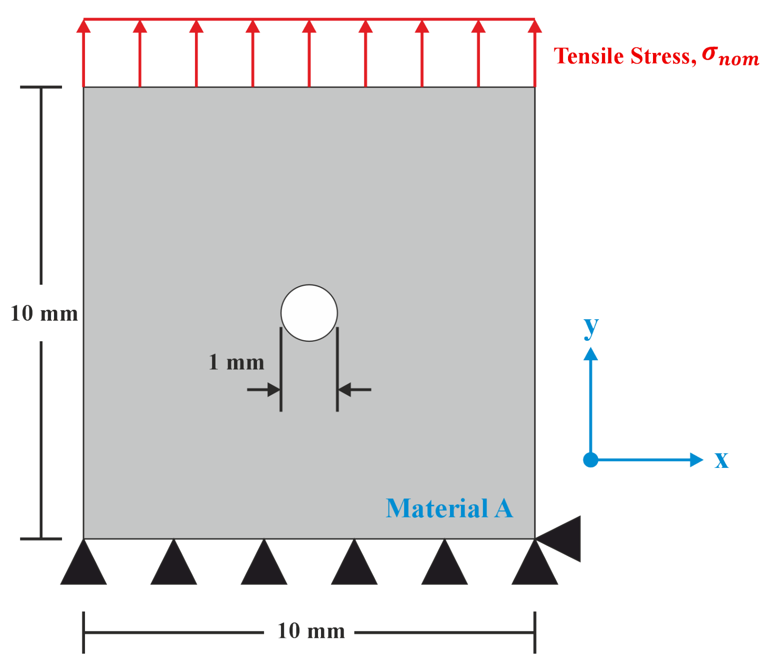


Figure 1: Baseline plate geometry with defined material A with thickness t = 1 mm. Two boundary conditions are applied, which are the fixed support restriction in y direction on the bottom line of the plate, fixed support point on the bottom right point restriction to x direction, and stress constant distribution along the top plate line with value of $\sigma_{nominal}$ = 1 MPa

2.2 Material Properties

The material properties used in the analysis are provided in the task definition. However, some parameters must be manually calculated based on the author's Immatrikulation Nummer in the very first section.

Table 1: Elastic properties for Q1.

E (GPa)	ν (Poisson's Ratio)
240	0.3

Table 2: Material parameters (combined hardening law) for Q2.

σ_y (MPa)	C_1 (MPa)	γ_1	C_2 (MPa)	γ_2	Q-Infinity	Hardening Param b
732	19996	44	3664	0	440	20

Table 3: Plastic properties for Q3

σ_{ya} (MPa)	σ_f (MPa)	ϵ_p^f
360	550	0.4

Table 1 shows the elastic properties that will be used in Question 1. These properties consist of Young's modulus (E) and Poisson's ratio (ν). Table 2 summarizes the material parameters for the combined hardening law used in Question 2, whereas some of the feature will be deactivated, for instance to model the isotropic hardening, the kinematic hardening parameters will be set to zero, and vice versa. Table 3 presents the plastic properties for question 3, which will be used in both the standard J2 plasticity model and the UMAT subroutine implementation.

Results and Discussion

Q1: Mesh Sensitivity Analysis of plate with circular hole

... Create the model as shown in figure 1, using the face partitioning illustrated in figure 2, and apply the following assumptions: small deformations, plane stress conditions, and a homogeneous, isotropic, linear elastic material. For each mesh size specified in table 1, create a separate Job. Apply the local mesh size around the hole and use the global mesh size for the remaining regions of the plate.

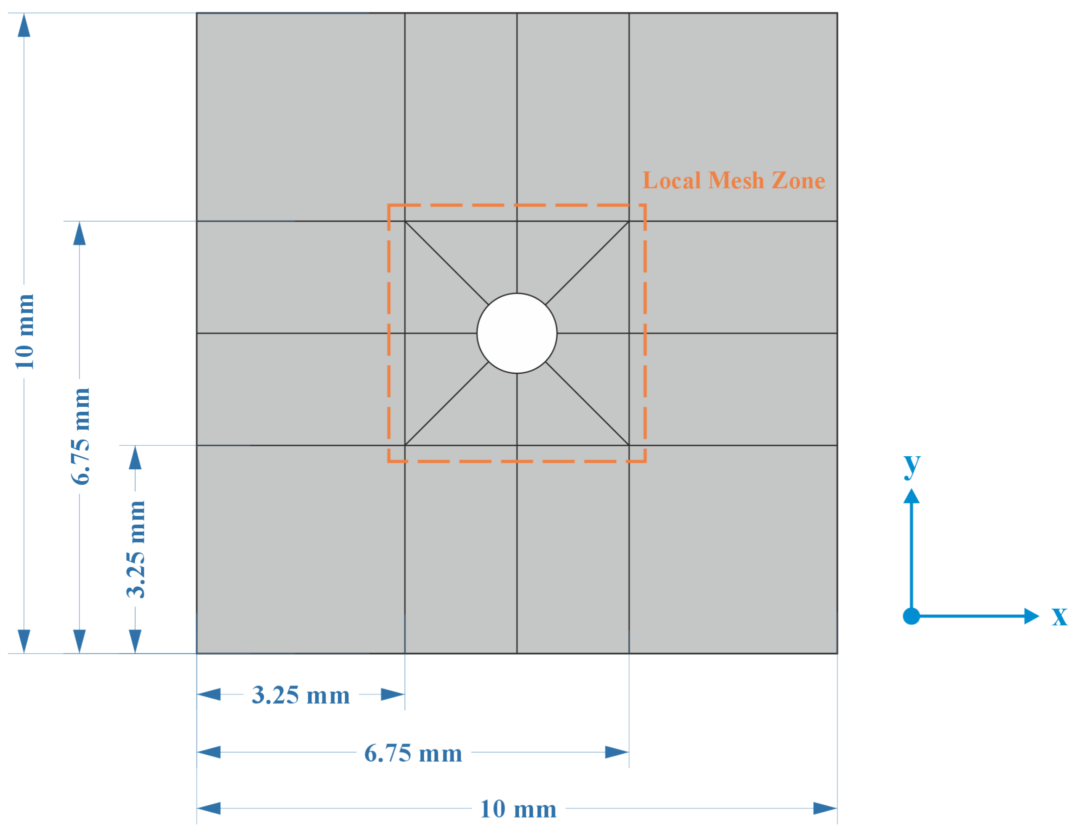


Figure 2: Partitioning guideline in order to create a structured mesh for the plate with circular hole. The whole mesh variations use CPS4, which is a 4-node bilinear plane stress quadrilateral element, also using quad-structured mesh control feature for the whole region to retain structured mesh and avoid extreme-distorted element.

The baseplate with circular hole is mesh using a structured mesh, hence the partitioning is well-defined to ensure a good mesh quality. As requested by the task, the baseplate is partitioned using the general formatting guidelines. However, the author have the freedom to adjust the partitioned size. Therefore, for this assignment the dimensioning of the mesh partition follows the provided Figure 2.

After the plate partitioning, specifically for task 1, the baseplate will be mesh using 8 variations in the mesh size, where it is based on the global and local mesh size. The global mesh size is defined as the mesh size for the entire plate, while the local mesh size is defined as the mesh size around the circular hole (hence is the area inside the small square partition). This method is done in order to capture the stress concentration effects on the area near the hole accurately while maintaining a low computational effort since the mesh size is not fine for the whole plate.

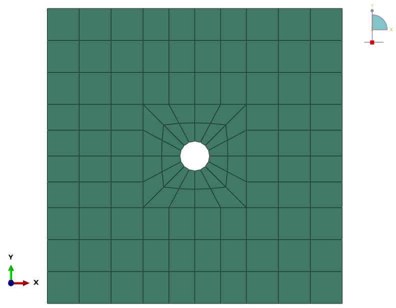
In order to know the mesh influence on the results, a mesh sensitivity analysis is performed. This analysis involves systematically varying the mesh sizes and observing the effects on the simulation results, particularly the stress distribution around the circular hole. Therefore, mesh variation in table 4 is used for this study, with a different global and local mesh size.

Table 4: Mesh size variations used for mesh sensitivity analysis.

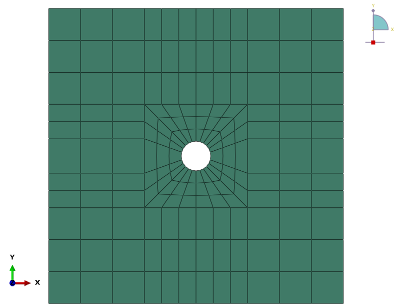
Mesh Number	Global Mesh Size (mm)	Local Mesh Size (mm)
1	1	1
2	1	0.6
3	0.8	0.4
4	0.6	0.3
5	0.4	0.2
6	0.3	0.1
7	0.2	0.05
8	0.1	0.02

Table Description: This table summarizes the eight mesh configurations used in the mesh sensitivity analysis. Each mesh is defined by its global mesh size (applied to the entire plate) and a finer local mesh size (applied around the circular hole) to accurately capture stress concentrations while optimizing computational efficiency.

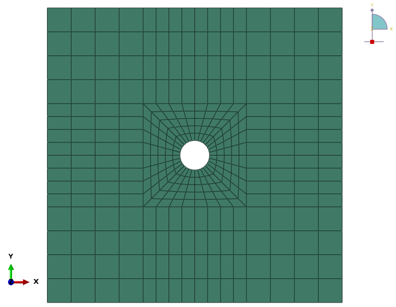
... For each mesh size, provide a figure exported directly from the Abaqus software (not a screenshot of the interface).



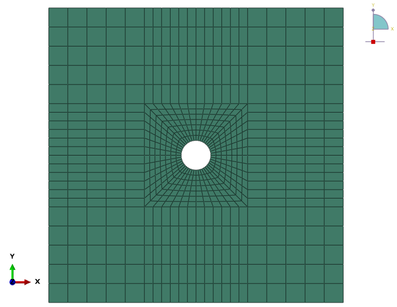
Mesh 1: 116 elements.



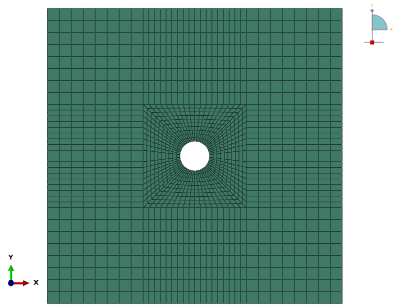
Mesh 2: 180 elements.



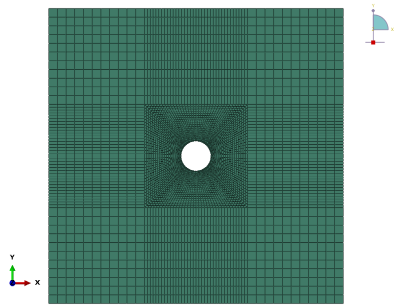
Mesh 3: 352 elements.



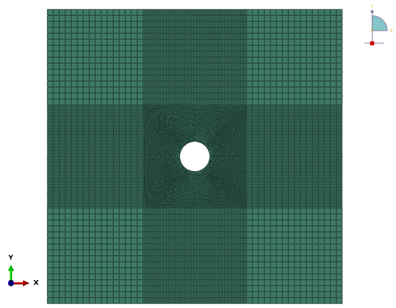
Mesh 4: 676 elements.



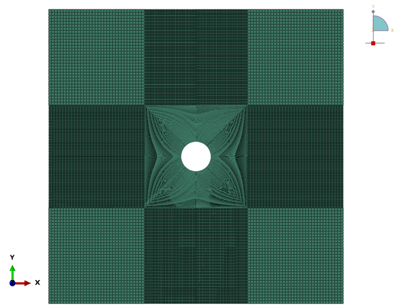
Mesh 5: 1552 elements.



Mesh 6: 4948 elements.

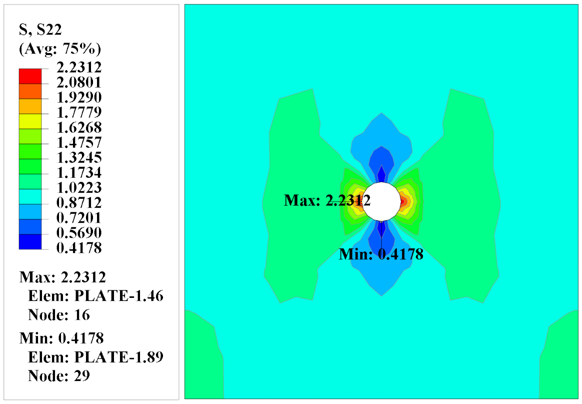


Mesh 7: 16424 elements.

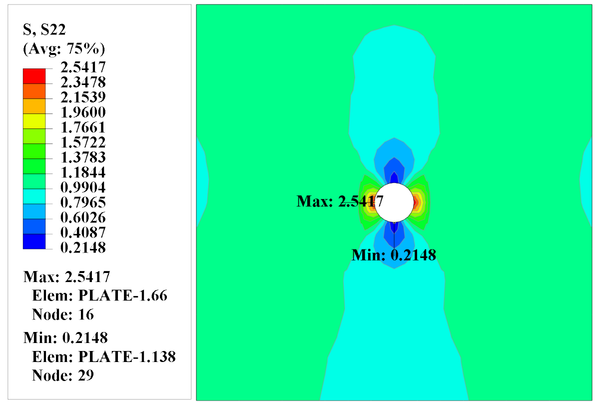


Mesh 8: 97042 elements.

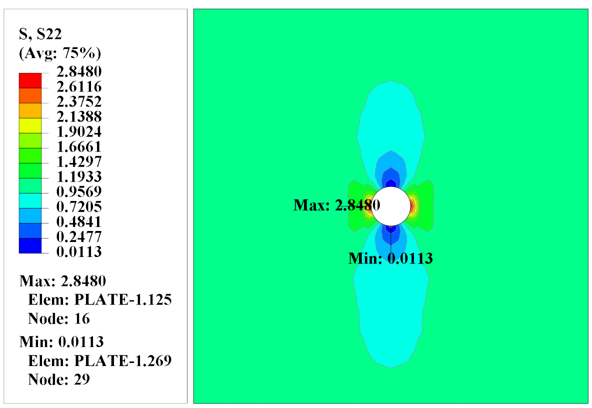
... For each mesh size, provide a contour plot of the relevant stress component, with indicating the locations of both maximum and minimum stress values on the plot. Please ensure the contour plots are clearly presented, and the values are readable.



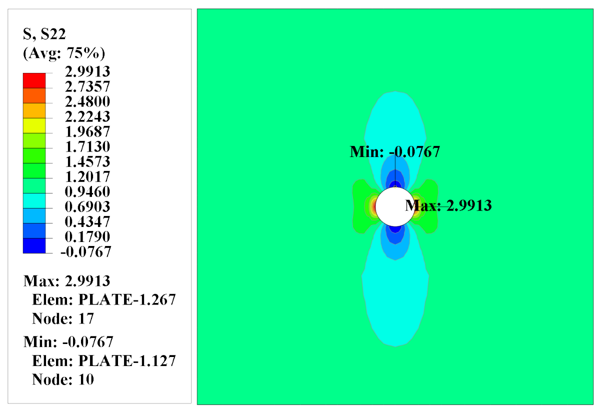
Mesh 1



Mesh 2



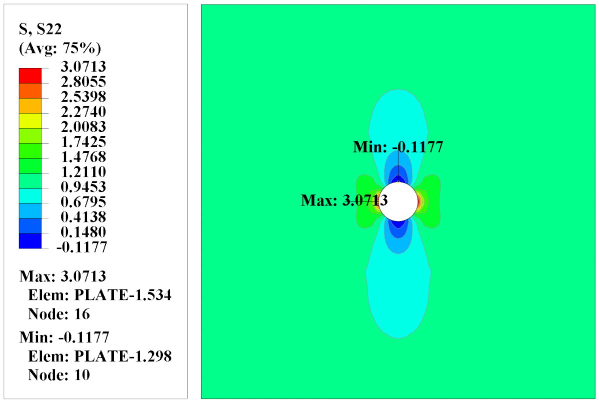
Mesh 3



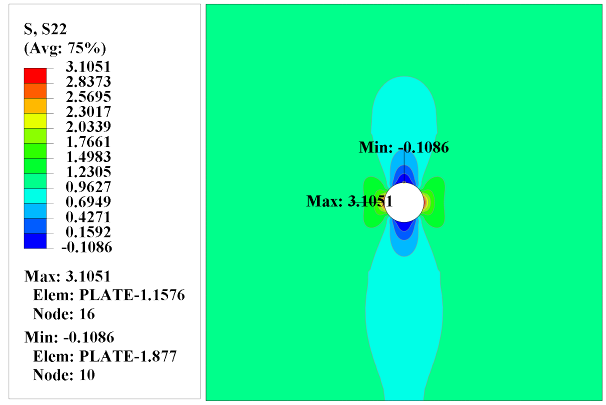
Mesh 4

Figure 3: Contour plots of the S_{22} stress component for Mesh 1, 2, 3, and 4 (from left to right, top to bottom). Maximum and minimum stress locations are indicated.

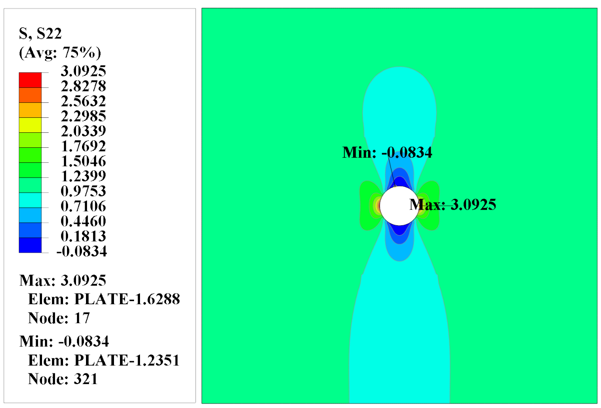
The relevant stress component for this analysis is the S_{22} stress component, which corresponds to the vertical stress in the y-direction. This component is particularly important for this case because the applied load is a uniaxial tensile stress in the vertical direction. Figure 3 shows the contour plots of the S_{22} stress component for the first four mesh sizes. A coarse stress distribution is shown in Mesh 1, with a maximum stress of 2.231 MPa. As the mesh is refined, the maximum area (hence refer to the red area) becomes more localized around the hole, indicating a more accurate capture of the stress concentration effect.



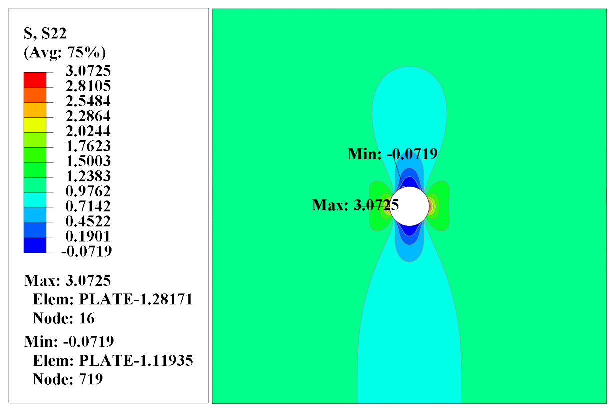
Mesh 5



Mesh 6



Mesh 7



Mesh 8

Figure 4: Contour plots of the S_{22} stress component for Mesh 5, 6, 7, and 8 (from left to right, top to bottom). Maximum and minimum stress locations are indicated.

As the mesh variation increase up to 1000 elements starting from Mesh 5 variation, the stress distribution becomes more refined. However, there are some enlargement of positive stress below the hole as the mesh is getting finer. This phenomenon can occur due to the interaction between the mesh refinement and the stress gradient around the hole.

... For each mesh size, calculate the stress concentration factor $K_t = \sigma_{max}/\sigma_{nom}$*, You are required to provide two plots: the first plot should show eight curves (one for each mesh size), the relevant stress over the path. The second plot should display the mesh number against* K_t*. Based on the results, discuss and justify which mesh provides the most accurate and efficient representation of stress concentration, and explain why it should be chosen.*

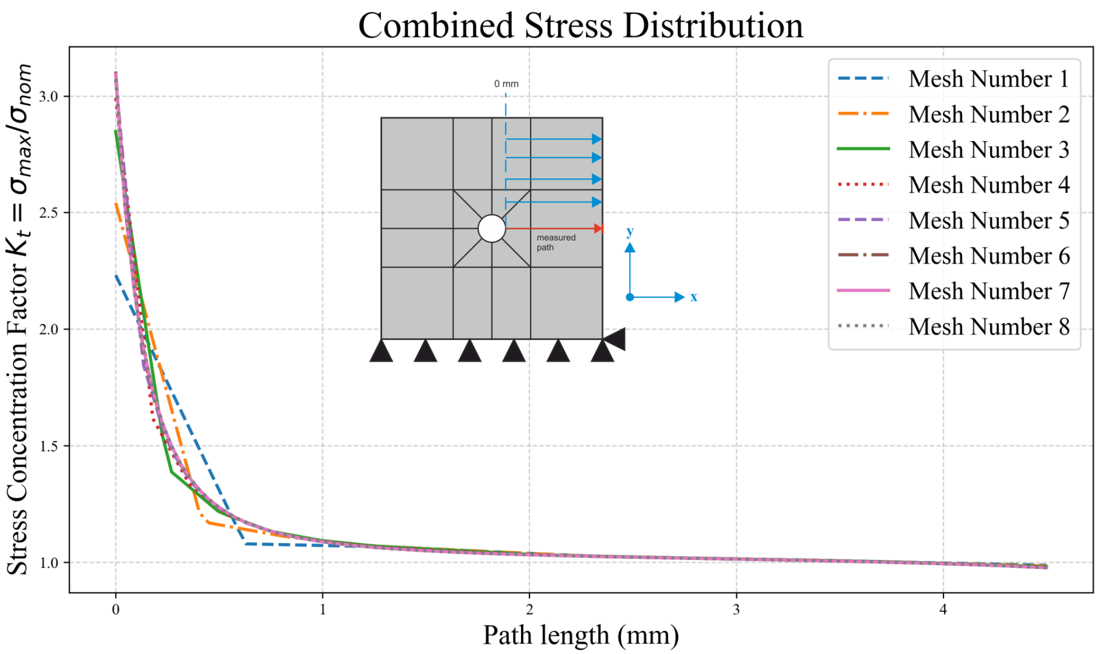


Figure 5: Stress concentration distribution along the defined path for all mesh sizes. Each curve represents the relevant stress component for a specific mesh configuration, illustrating the effect of mesh refinement on capturing the stress concentration near the circular hole. The path is defined from the edge of the hole to the opposite edge of the plate, marked by redline in the geometry. The stress concentration factor is calculated based on the ratio of generated stress or maximum stress in the specific point to the nominal stress, which is 1 MPa.

Figure 5 shows the stress distribution of the S_{22} stress component along the defined path for all mesh sizes. As the mesh is refined, there are some tendencies that appear in the development of the stress distribution curve. The first tendency is the maximum stress value is increasing as the mesh is getting finer. This is due to the fact that a finer mesh can capture the stress concentration more accurately, leading to higher peak stress values. However, after Mesh 6 variation, the maximum stress value starts to decrease slightly as shown in figure 6. This indicates that the solution is converging, and further mesh refinement does not significantly change the results.

The second tendency is the overall shape of the stress distribution curve becomes smoother as the element size decreases. This is because a finer mesh has the ability to capture the stress gradients more accurately, resulting in a more continuous stress curves along the path.

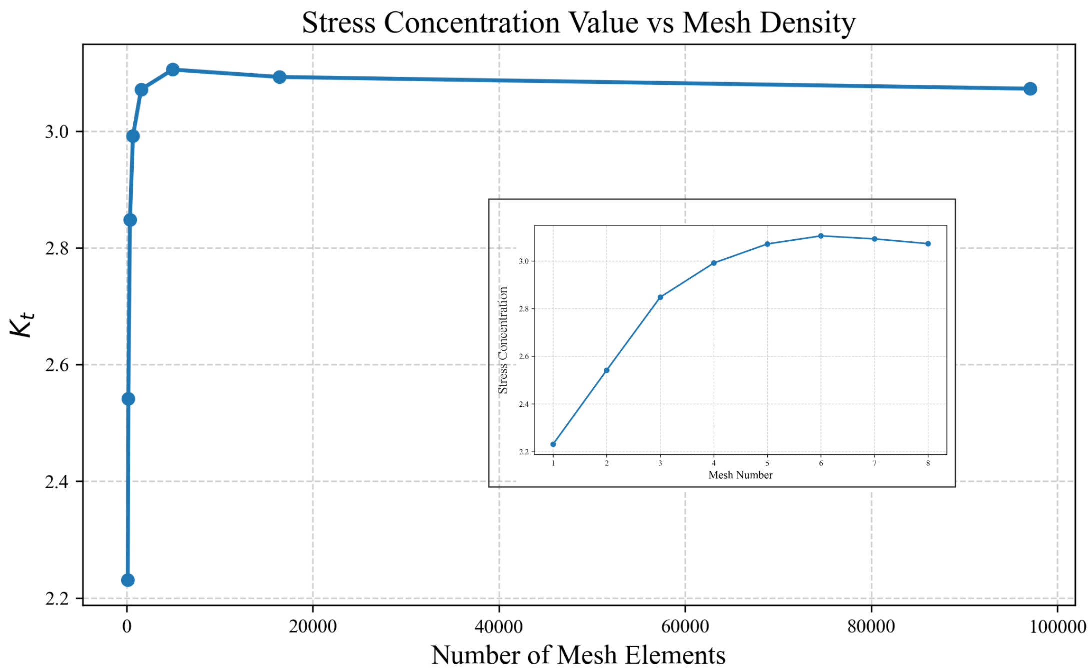


Figure 6: Stress concentration factor K_t along the defined path for all mesh sizes. Each curve represents the K_t value for a specific mesh configuration, illustrating the effect of mesh refinement on capturing the stress concentration near the circular hole. The plot is divided by two plots, where the main plot shows the overall K_t value based on the element numbers that being used, while the zoomed-in plot shows the detail of K_t value based on specifically the mesh numbering.

Based on these results, mesh number 4 is chosen as the most optimal mesh sizing configuration. As we refer to the classical value of K_t, which is 3.0 for plate structures, we can see that mesh 4 provides a K_t value that is in close agreement with this classical solution. This indicates that the mesh is sufficiently refined to capture the stress concentration effects accurately without being overly computationally expensive. It is possible that to use more refined mesh than mesh 8, since the curve is about to converge to value of 3.0. However, the computational cost will become significantly higher, while the improvement in accuracy will not be substantial. Therefore, mesh 4 is selected as the optimal mesh for this analysis, balancing accuracy and efficiency.

... Taking advantage of the symmetry in both the geometry and the boundary conditions, propose a simplified model that reduces computational cost and simulation time. Create and submit two figures: one showing the mesh of the proposed model, and the other displaying contour plot of the relevant stress distribution. Compare it to the original model.

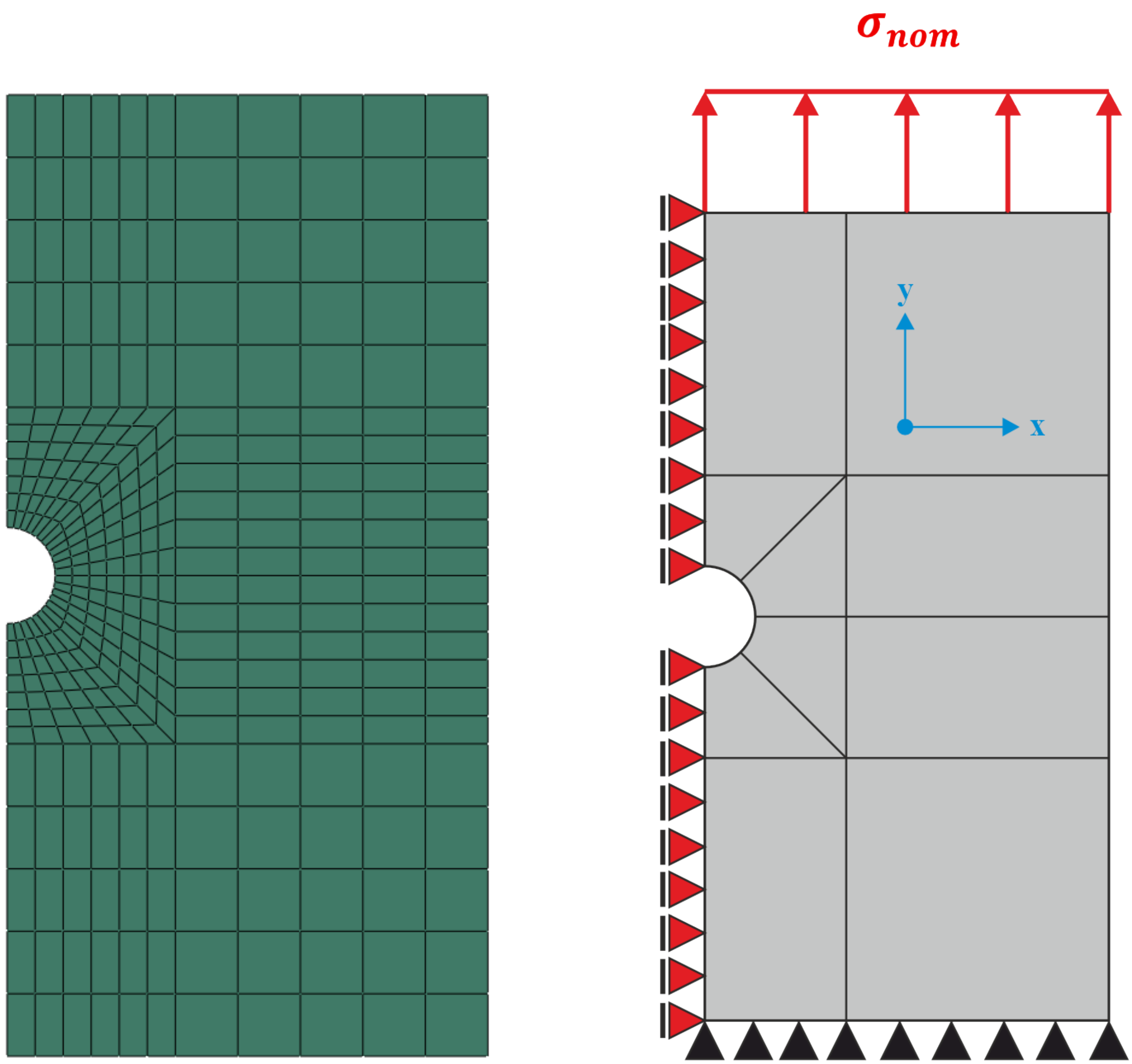
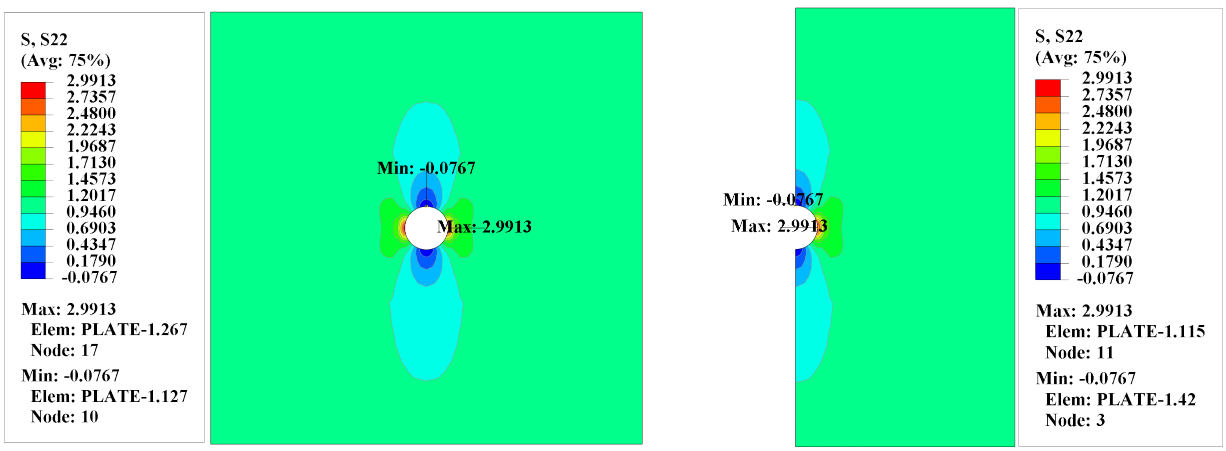


Figure 7: (Left) Mesh of the proposed half-symmetry model. (Right) boundary conditions configuration applied to the half model. Additional boundary condition is applied to the symmetry line, which here restrict the horizontal movement while allowing y-axis movement. The restriction on the right bottom corner is omitted since it is already restricted by the symmetry line for the horizontal movement. The nominal stress is applied on the top line as in the full model, as well for the fixed support on the bottom that allows horizontal movement.

The symmetry is modeled as in figure 7, where the mesh is created using the same mesh size as in Mesh 4 variation. Modification of boundary condition is done as in figure 7 so that the symmetry model can represent the full model. After running the simulation, the stress contour plot is shown in figure 8. It can be seen that the stress distribution for the symmetry model is identical with the full model. The value of maximum stress and minimum stress is also the same for both cases. Hence by utilizing symmetry, the computational cost and simulation time can be significantly reduced by calculating fewer areas which represent the full model.



Full model

Half-symmetry model

Figure 8: Comparison of S_{22} stress contours for Mesh 4: full model (left) and half-symmetry model (right). Both plots show the stress distribution and concentration near the hole, demonstrating that the half-symmetry model accurately reproduces the results of the full model.

Q2: Investigate the effects of isotropic and kinematic hardening on the mechanical behavior of a material

In this question, the focus will be to investigate the isotropic and kinematic hardening effect to its mechanical response. The analysis will be conducted using an uniaxial tensile test model, where the geometry is from previous question 1. But here instead of using a nominal stress $\sigma_{nominal}$, a displacement of 0.2 mm will be applied to the top edge of the plate, while the rest of the boundary conditions is similar as in figure 2. However, since hardening effects would be investigated, cyclic loading will be applied to the model. A cyclic amplitude in figure 9 is used to perform the cyclic loading simulation based on the total displacement of 0.2 mm.

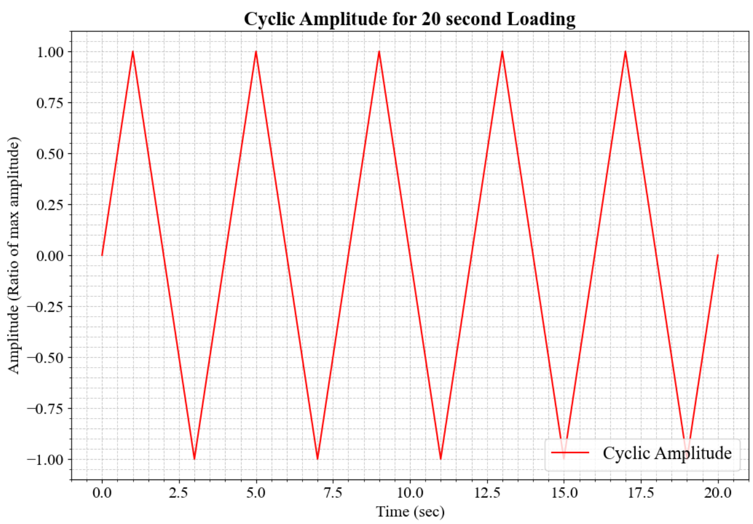


Figure 9: Applied cyclic loading amplitude for the tension-compression simulation. The plot shows the displacement history used to investigate the hardening behavior under repeated loading cycles.

...***Isotropic Hardening Model:*** *Set the material parameters for the isotropic hardening model, keeping kinematic hardening deactivated. Perform a cyclic loading simulation (tension-compression) using the amplitude plotted in figure 3. Plot and analyze the stress-strain hysteresis loops.*

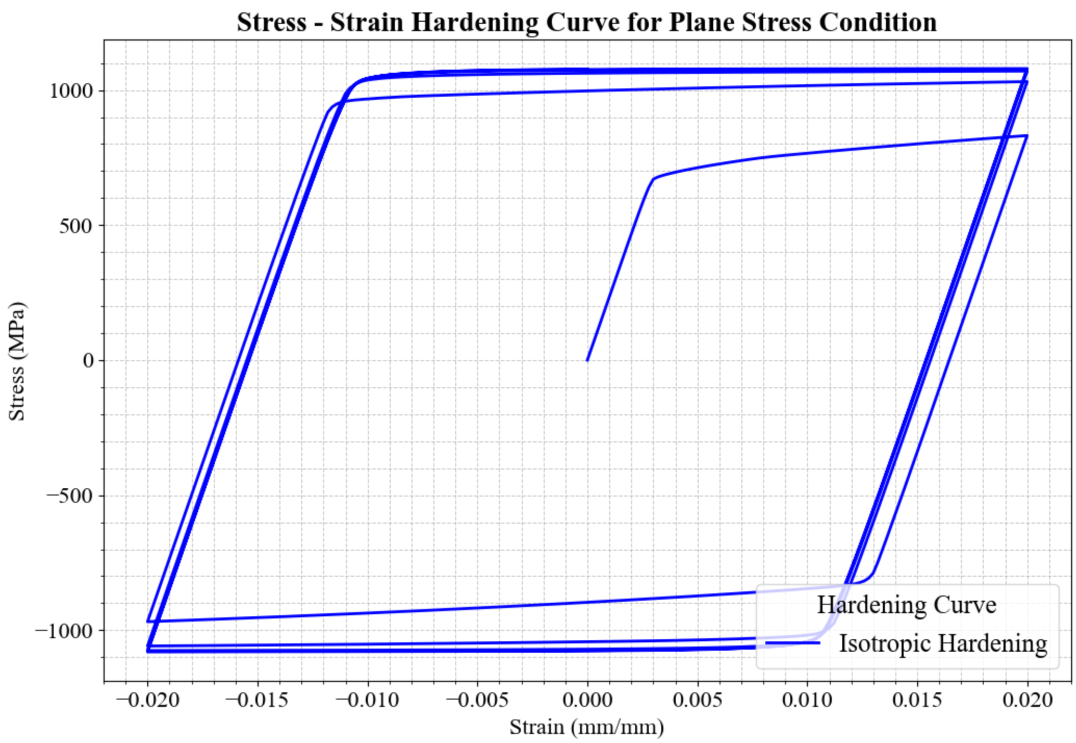


Figure 10: Stress-strain hysteresis loop for the isotropic hardening model under cyclic tension-compression loading. The plot illustrates the expansion of the yield surface and the corresponding material response. For this isotropic hardening model, Q-infinity and hardening parameter b are set according to 2, whereas the C_1, γ_1, and C_2 are set to zero to disable the kinematic hardening effect.

Figure 10 shows the stress-strain hysteresis loop for the isotropic hardening model under cyclic tension-compression loading. As the cyclic load applied, at first cycle the material starts to yield at the initial yield stress of 732 MPa. As the loading continues and changing direction to compression, the yield surface starts to expand isotropically, leading to an increase in the yield stress for subsequent cycles. This expansion of the yield surface is characteristic of isotropic hardening, where the material's resistance to plastic deformation increases uniformly in all directions.

. . . ***Kinematic Hardening Model:*** *Set the material parameters for the kinematic hardening model, assuming no isotropic hardening. Perform a cyclic loading simulation (tension-compression) using the amplitude plotted in figure 3. Plot and analyze the stress-strain hysteresis loops.*

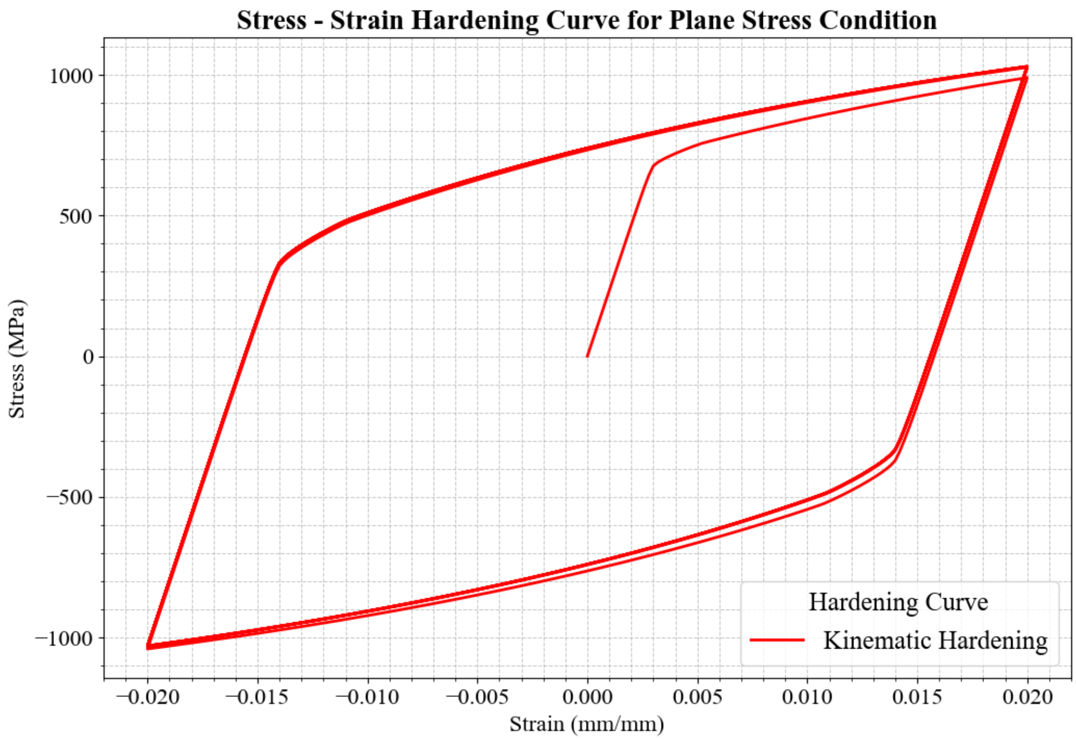


Figure 11: Stress-strain hysteresis loop for the kinematic hardening model under cyclic tension-compression loading. The plot demonstrates the translation of the yield surface and the corresponding material response.

Figure 11 shows the stress-strain hysteresis loop for the kinematic hardening model under cyclic tension-compression loading. Unlike the isotropic hardening case, the yield surface translates in the stress space without changing its size. This behavior indicates that kinematic hardening occurs, where the material's yield surface moves in response to plastic deformation, allowing for a more accurate representation of the Bauschinger effect.

. . . *Compare the extracted curves from previous questions and discuss the results.*

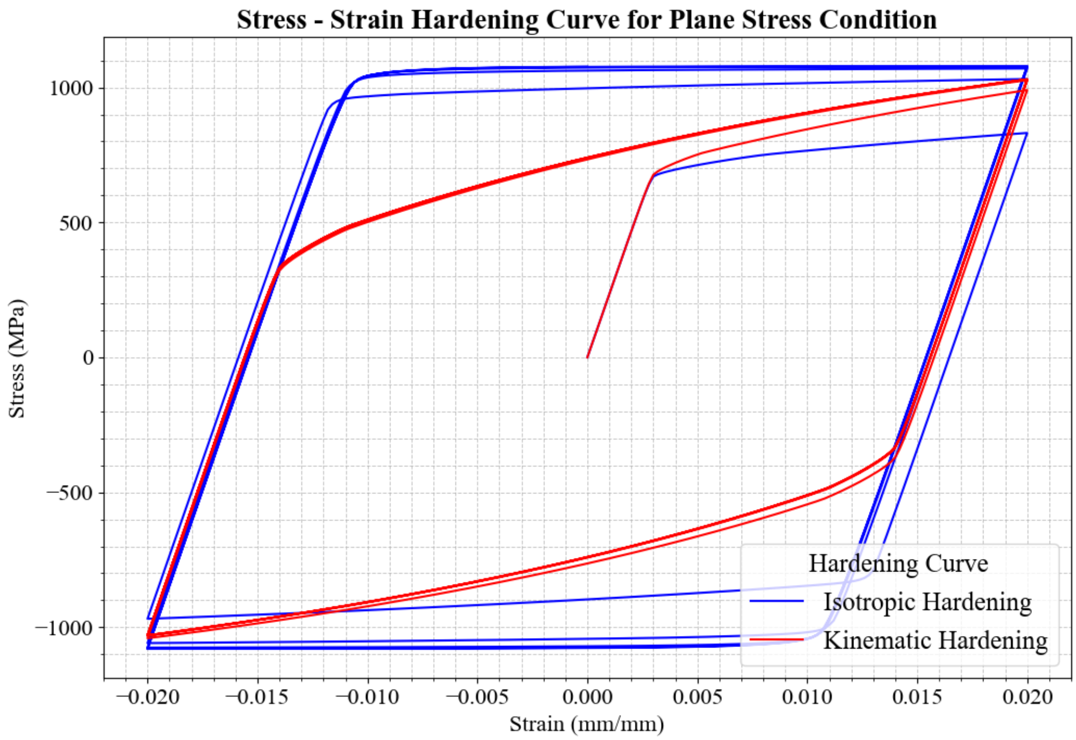


Figure 12: Comparison of stress-strain hysteresis loops for isotropic and kinematic hardening models under cyclic tension-compression loading. The plot highlights the differences in material response due to the various hardening mechanisms.

Figure 12 shows the comparison of stress-strain hysteresis loops for isotropic and kinematic hardening models under cyclic tension-compression loading. The isotropic hardening model exhibits a symmetric hysteresis loop, indicating uniform expansion of the yield surface in all directions. In contrast, the kinematic hardening model shows an asymmetric hysteresis loop, reflecting the translation of the yield surface in response to plastic deformation. The difference from these two models is obvious, where the isotropic hardening model shows a larger area within the hysteresis loop, indicating higher energy dissipation during cyclic loading. This is due to the uniform increase in yield stress that allows in this model. While in the kinematic hardening model, increase in yield stress does not occur, only shifting of yield surface is allowed. Therefore, the energy dissipation is lower compared to the isotropic hardening model.

. . . Simulate the model using combined hardening law parameters in table 2. Compare the results with those obtained using isotropic and kinematic hardening parameters. Discuss how variations in the combined hardening parameters influence the material behavior.

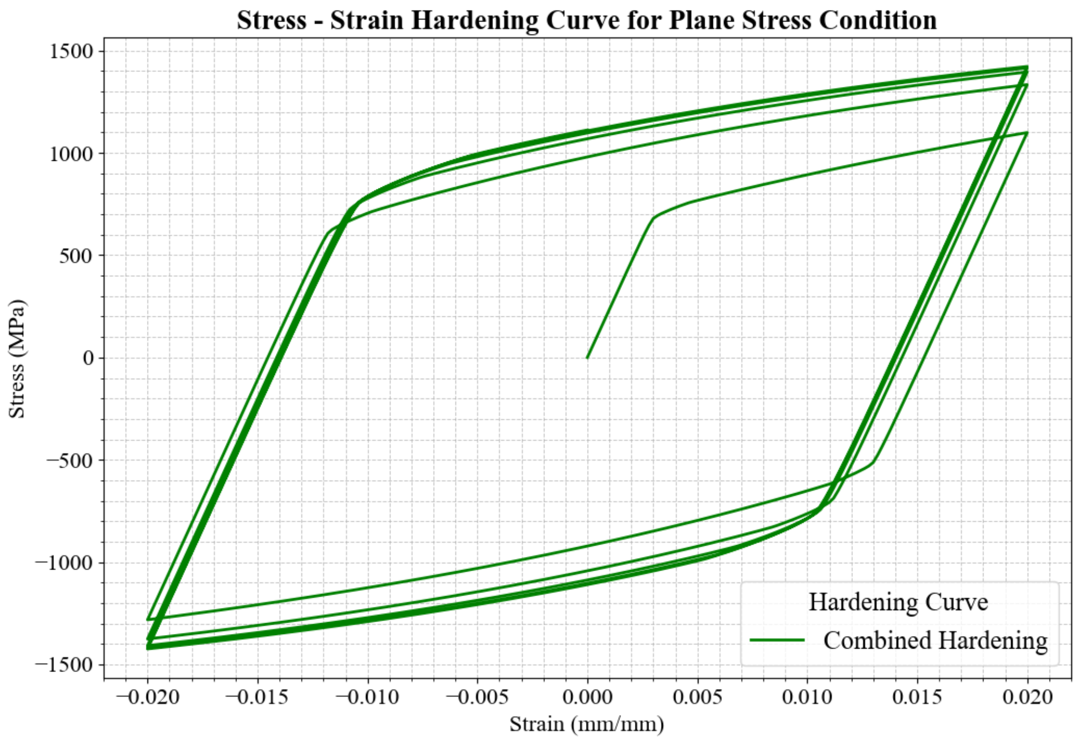


Figure 13: Stress-strain hysteresis loop for the combined hardening model under cyclic tension-compression loading. The plot shows the material response when both isotropic and kinematic hardening mechanisms are active, illustrating the combined effects on the expansion and translation of the yield surface.

As both of the model are combined, the stress-strain hysteresis loop for the combined hardening model is shown in figure 13. This shows a unique material behavior, whereas the cyclic loading is applied, the yield surface expands isotropically as well as translating in the stress space. This combined effect results in a more complex hysteresis loop, which reflects the play role between isotropic and kinematic hardening mechanisms. The combined hardening model produced a larger hysteresis loop area compared to the kinematic, as well as for the isotropic hardening model.

This is because the combined hardening model incorporates both of the hardening model, so that it have the ability to increase the yield stress surface as well shifting the yield surface. Therefore, the energy dissipation during cyclic loading is higher than the other two models. A comparison of all three hardening models is shown in figure 14.

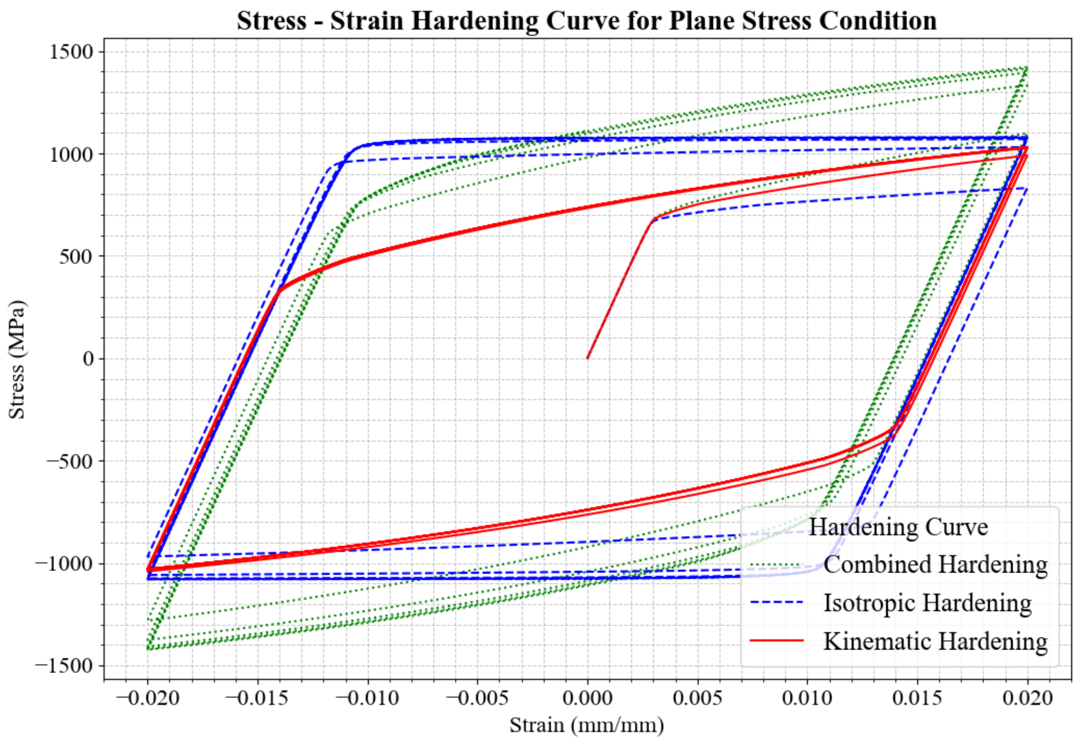


Figure 14: Comparison of stress-strain hysteresis loops for isotropic, kinematic, and combined hardening models under cyclic tension-compression loading. The plot highlights the differences in material response due to the various hardening mechanisms.

Q3: Standard and UMAT subroutine for J2 plasticity

... Read the provided J2 UMAT, then decide to choose plane stress or plane strain condition, add a screenshot that shows the code lines where this choice is made and where elastic stiffness and equivalent von mises stress is calculated, and give an explanation based on the lecture.

```
C
C ---------------------------------------------------------------
C UMAT FOR ISOTROPIC ELASTICITY AND ISOTROPIC MISES PLASTICITY
C CANNOT BE USED FOR PLANE STRESS
C
C PROPS(1) - E
C PROPS(2) - NU
C PROPS(3...) - YIELD AND HARDENING DATA
C CALLS UHARD FOR CURVE OF YIELD STRESS VS. PLASTIC STRAIN
C ---------------------------------------------------------------
```

Figure 15: Code section in the UMAT where the plane strain or plane stress condition is selected. This determines the appropriate constitutive behavior for the simulation.

```
C    ELASTIC STIFFNESS
C
      DO K1=1, NDI                    ! -      -    -                                    -  -       -
        DO K2=1, NDI                  ! |sigma11|  |2G+Lambda              Sym.          ||strain11 |
          DDSDDE(K2, K1)=ELAM         ! |sigma22|  | Lambda    2G+Lambda                 ||strain22 |
        END DO                        ! |sigma33|= | Lambda      Lambda   2G+Lambda      ||strain33 |
        DDSDDE(K1, K1)=(EG2+ELAM)     ! |sigma12|  |   0           0          0      G   ||2strain12|
      END DO                          ! |sigma13|  |   0           0          0      0  G  ||2strain13|
      DO K1=NDI+1, NTENS              ! |sigma23|  |   0           0          0      0  0 G||2strain23|
        DDSDDE(K1, K1)=EG             ! -      -    -                                    -  -       -
      END DO
```

Figure 16: Code lines in the UMAT where the elastic stiffness matrix is calculated. This matrix defines the linear elastic response of the material.

```
C CALCULATE EQUIVALENT VON MISES STRESS
C
      SMISES=(STRESS(1)-STRESS(2))**2+(STRESS(2)-STRESS(3))**2
     1       +(STRESS(3)-STRESS(1))**2
      DO K1=NDI+1,NTENS                     !          /-------------------------------------------------------------------------------------------
        SMISES=SMISES+SIX*STRESS(K1)**2     !sigma_e=\/ 1/2[(sigma11-sigma22)^2+(sigma33-sigma22)^2+(sigma11-sigma33)^2+6(sigma12^2+sigma13^2+sigma23^2)]
      END DO
      SMISES=SQRT(SMISES/TWO)
```

Figure 17: Code section in the UMAT where the equivalent von Mises stress is computed. This value is used to evaluate yielding according to the J2 plasticity criterion.

References to these figures can be made as Figure 15, Figure 16, and Figure 17.

... Run two simulations with a displacement of 0.5 mm and same material properties: one with the Abaqus embedded J2 model and one with UMAT. Plot, compare, and discuss the true stress-strain curve for both cases.

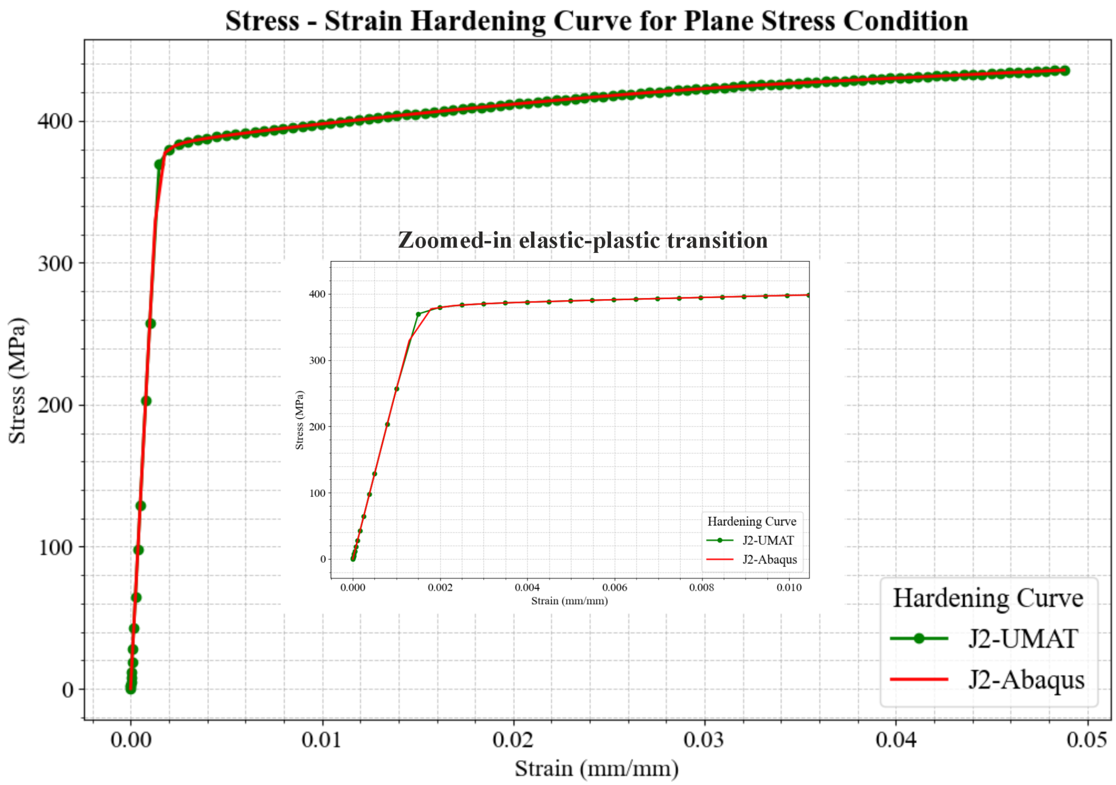
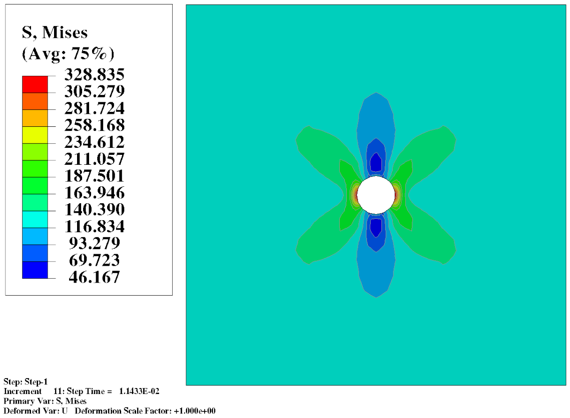


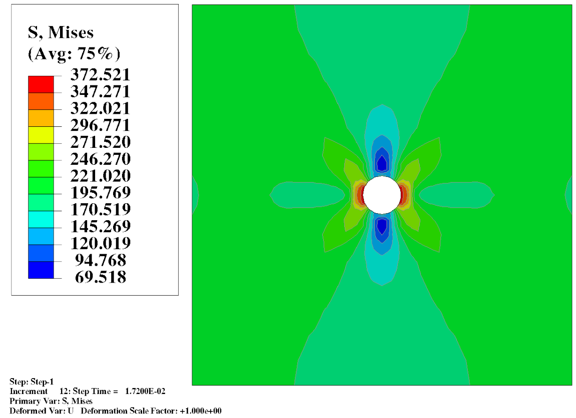
Figure 18: Zoomed-in view of the true stress-strain curve from the tensile simulation, highlighting the detailed behavior near the area of necking and plastic deformation.

Figure above shows the true stress-strain curves for J2-UMAT model and J2-Abaqus. Both of the curve having the same overall behavior in the elastic and plastic area. However in the elastic-plastic transition area, J2-Abaqus yielded first by showing the slope changing in the curve, while J2-UMAT still maintain its slope at the same level of strain value. From this it can be concluded that the J2-UMAT model exhibits a more stable response during the transition from elastic to plastic behavior. This shows by a higher value for the yielding point in J2-UMAT and the change to plastic region is more sudden rather gradual compare to the J2-Abaqus model.

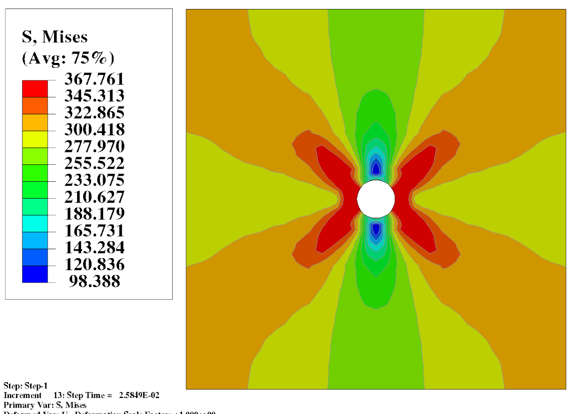
. . . From the embedded J2 simulation, select four representative time frames that trace the evolution from the purely elastic status to the onset of localized necking, present for each frame a contour plot of von Mises stress and equivalent plastic strain. Describe the deformation process of the plate with a hole.



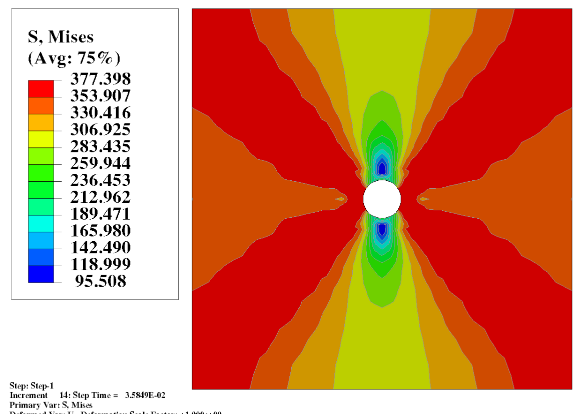
(a) von Mises stress, Frame 1



(b) von Mises stress, Frame 2

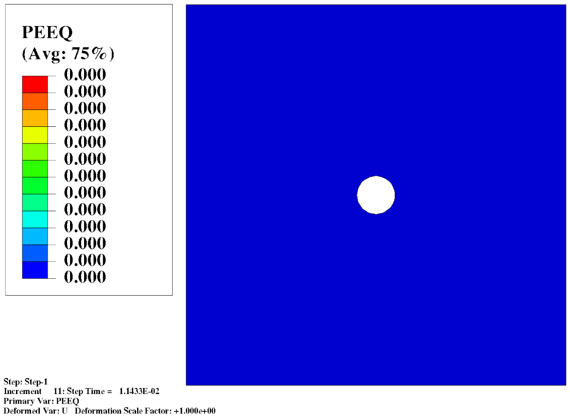


(c) von Mises stress, Frame 3

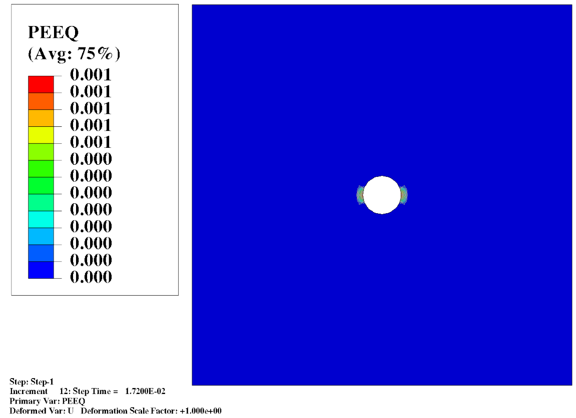


(d) von Mises stress, Frame 4

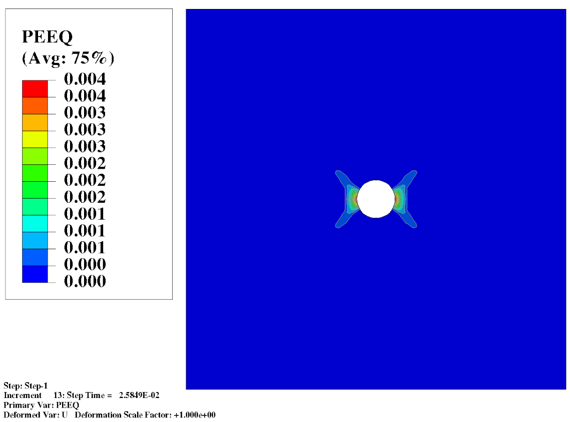
Figure 19: Contour plots of von Mises stress at four representative time frames from the embedded J2 simulation, showing the evolution from elastic to localized necking. The von Mises stress grows starting from the initial loading phase in time frame 1, where it is at the elastic region. At frame 2, it starts to deform plastically as the stress distribution starts to yield.



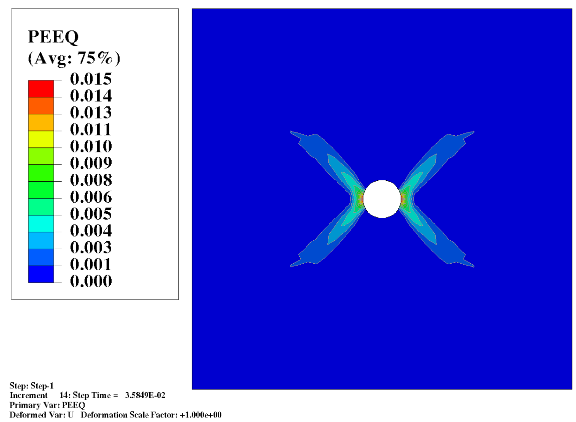
(a) Equivalent plastic strain, Frame 1



(b) Equivalent plastic strain, Frame 2



(c) Equivalent plastic strain, Frame 3



(d) Equivalent plastic strain, Frame 4

Figure 20: Contour plots of equivalent plastic strain at four representative time frames from the embedded J2 simulation, illustrating the progression of plastic deformation and localization.

. . . From the UMAT simulation, plot the distribution of the relevant stress and plastic strain components for the final time, indicating the meaning of the different SDVs, showing the maximum and minimum location, then discussing the results for different regions.

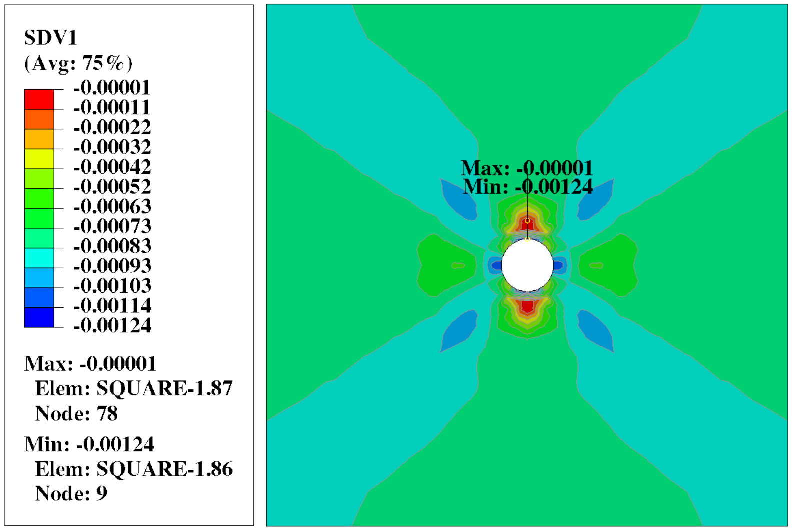


Figure 21: SDV1: Distribution of the first state-dependent variable (SDV1) at the final simulation time, representing the relevant stress or plastic strain component. Maximum and minimum locations are indicated.

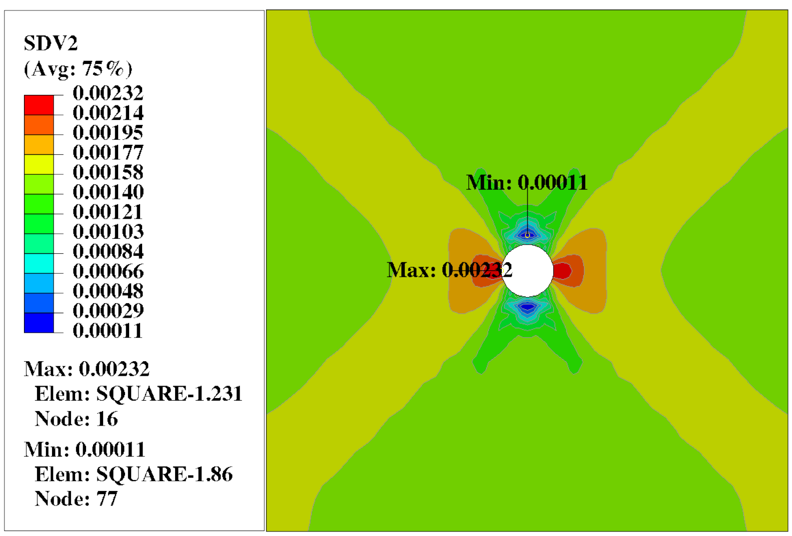


Figure 22: SDV2: Distribution of the second state-dependent variable (SDV2) at the final simulation time. This may correspond to an internal variable such as accumulated plastic strain or backstress.

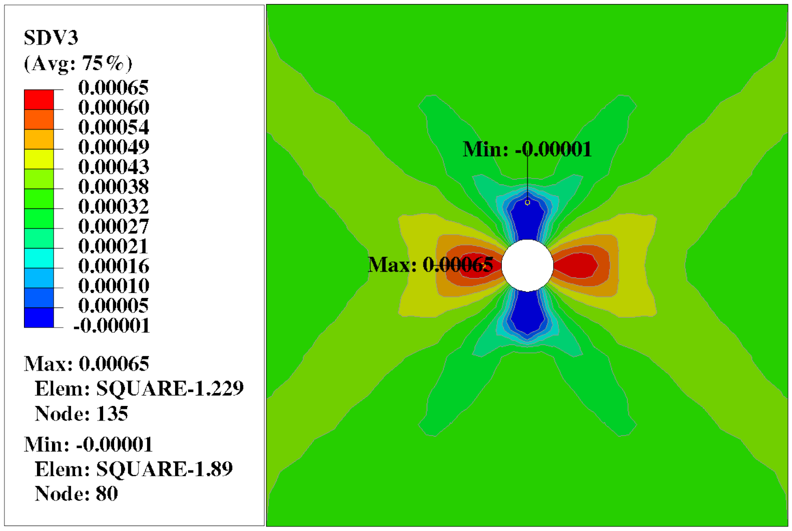


Figure 23: SDV3: Distribution of the third state-dependent variable (SDV3) at the final simulation time, indicating another relevant mechanical quantity tracked by the UMAT.

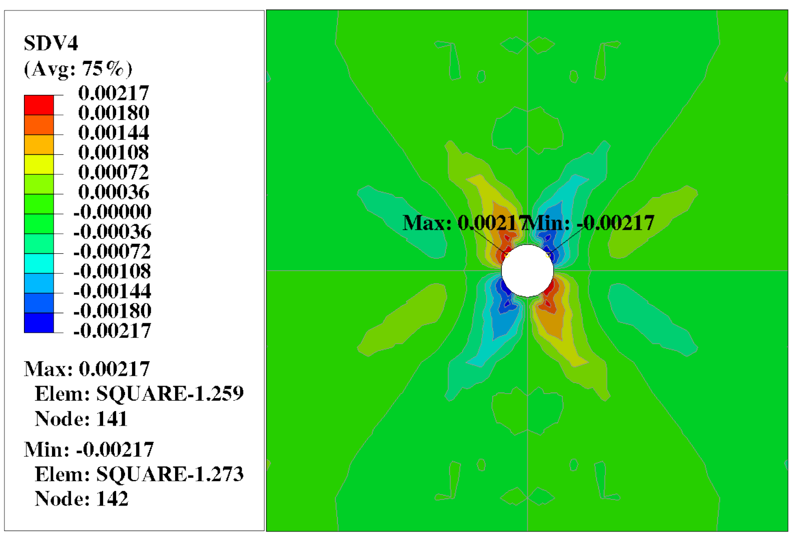


Figure 24: SDV4: Distribution of the fourth state-dependent variable (SDV4) at the final simulation time, showing the spatial variation and localization effects.

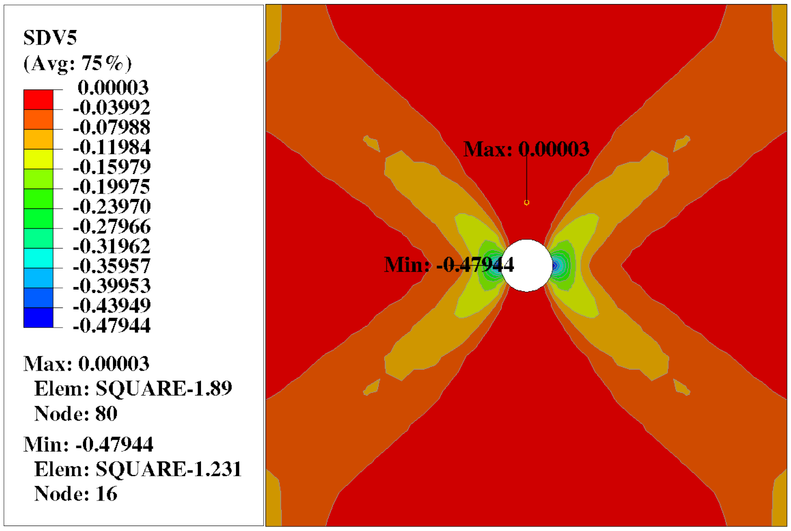


Figure 25: SDV5: Distribution of the fifth state-dependent variable (SDV5) at the final simulation time, highlighting regions of maximum and minimum values.

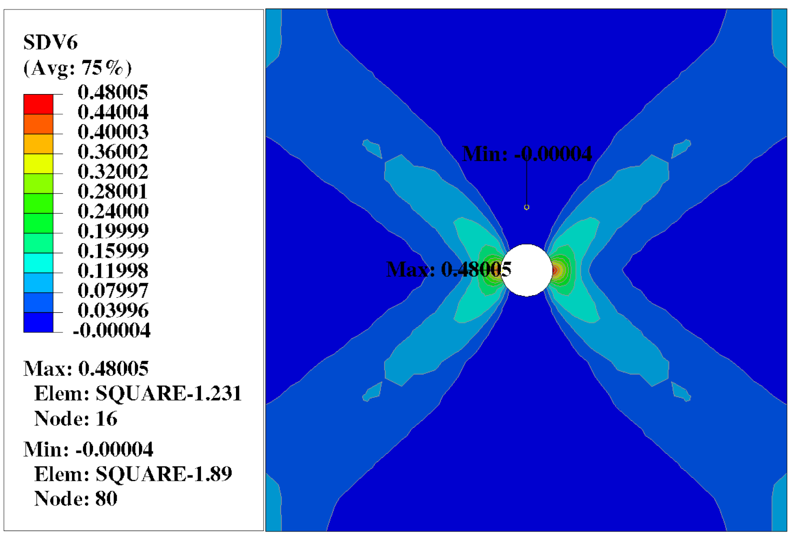


Figure 26: SDV6: Distribution of the sixth state-dependent variable (SDV6) at the final simulation time, illustrating the evolution of the corresponding mechanical response.

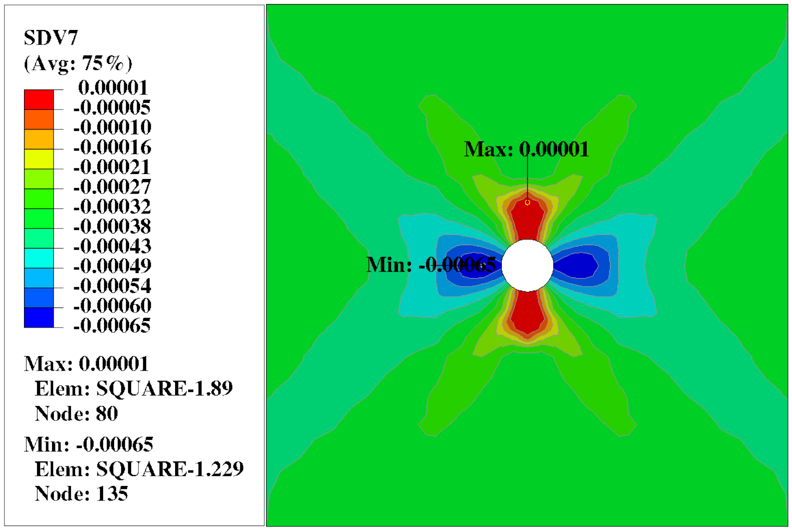


Figure 27: SDV7: Distribution of the seventh state-dependent variable (SDV7) at the final simulation time, with maximum and minimum locations marked.

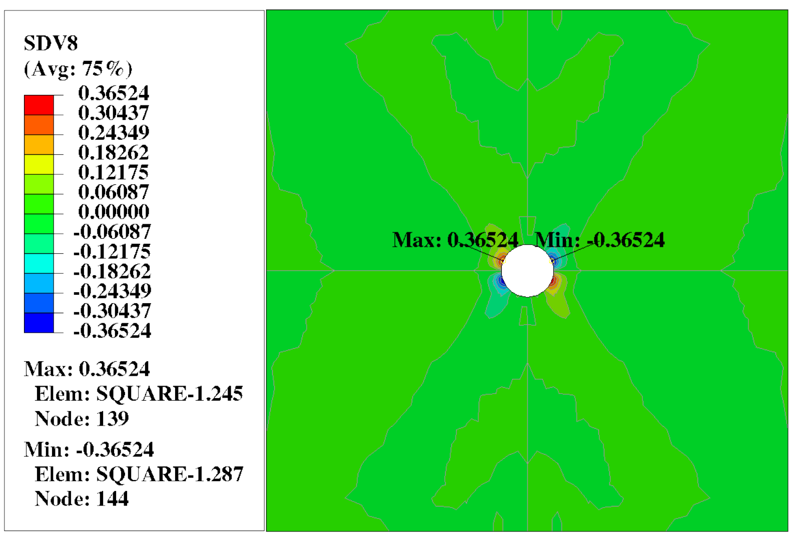


Figure 28: SDV8: Distribution of the eighth state-dependent variable (SDV8) at the final simulation time, showing the spatial variation across the plate.

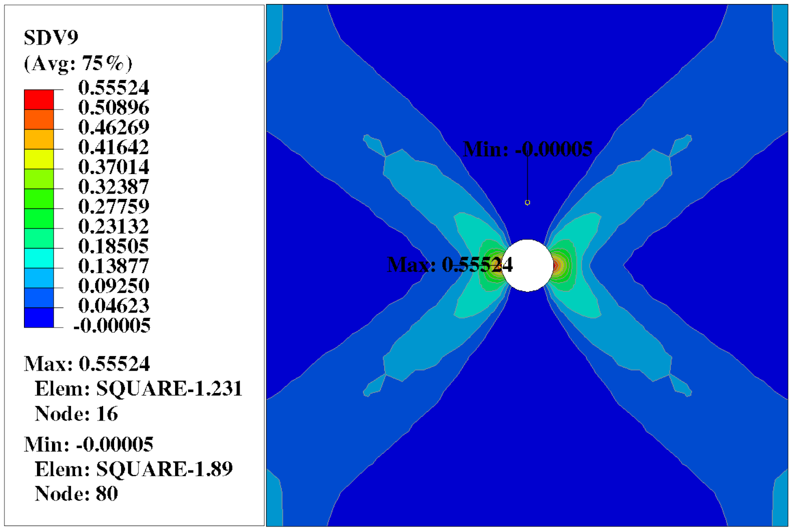


Figure 29: SDV9: Distribution of the ninth state-dependent variable (SDV9) at the final simulation time, indicating the final state of the relevant mechanical quantity.

Conclusion

To sum up everything that have been analyzed and gathered in this study, here are some several key takeaways from this small case study:

1.